

Question:

Kylar went to the store to buy glasses for his new apartment. One glass costs \$5, but every second glass costs only 60% of the price. Kylar wants to buy 16 glasses. How much does he need to pay for them?

Reasoning + Answer:

Let's break down the problem step by step to determine how much Kylar needs to pay for 16 glasses .

1 . **Understanding the pricing structure : **

- The cost of the first glass is \$ 5 .
- The cost of every second glass is 60 % of \$ 5 .

2 . **Calculating the price of the second glass : **

- To find 60 % of \$ 5 , we use the following calculation :

$$\begin{aligned} & \backslash[\\ & 0.60 \times 5 \end{aligned}$$

- Performing the multiplication :

$$\begin{aligned} & \backslash[\\ & 0.60 \times 5 = 3 \end{aligned}$$

- Therefore , the cost of every second glass is \$ 3 .

3 . **Determining the pattern for 16 glasses : **

- Kylar wants to buy 16 glasses .
- In this pattern , every second glass costs \$ 3 , starting from the second glass .
- We need to count the number of glasses that cost \$ 3 and the number that cost \$ 5 .

4 . **Counting the glasses : **

- The first glass costs \$ 5 .
- The glasses that follow are in pairs : the second glass , fourth glass , sixth glass , etc .
- We need to find how many such pairs exist in 16 glasses .

5 . **Calculating the number of pairs : **

- Since the glasses are in pairs , we divide 16 by 2 :

$$\begin{aligned} & \backslash[\\ & \frac{16}{2} = 8 \end{aligned}$$

- This means there are 8 pairs of glasses .

6 . **Calculating the total cost for the pairs : **

- Each pair consists of one glass at \$ 5 and one glass at \$ 3 .
- The total cost for one pair is :

$$\begin{aligned} & \backslash[\\ & 5 + 3 = 8 \end{aligned}$$

- Since there are 8 pairs , the total cost for all pairs is :

$$\begin{aligned} & \backslash[\\ & 8 \times 8 = 64 \end{aligned}$$

$$\backslash]$$

7 . **Adding the cost of the first glass : **

- The first glass costs \$ 5 .
- Adding this to the total cost of the pairs :

$$\begin{aligned} & \backslash[\\ & 64 + 5 = 69 \end{aligned}$$

$$\backslash]$$

Therefore , the total amount Kylar needs to pay for 16 glasses is :

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- 1 . Cost of first glass : \$ 5
- 2 . Cost of every second glass : \$ 3
- 3 . Number of pairs : $16 / 2 = 8$
- 4 . Total cost for pairs : $8 * (5 + 3) = 8 * 8 = 64$
- 5 . Total cost including first glass : $64 + 5 = 69$

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<answer> 69 </answer>